

Lab: Tracking Your Training Epochs

Objective

Experience neural network training firsthand — with your hands. You will learn a new crochet stitch you have never done before, systematically track your errors over multiple rows, plot your personal training curve, and compare it to a typical neural network learning curve. This lab makes the parallel between skill acquisition and weight optimization tangible and personal.

Materials

- Worsted weight yarn in a light color (for visibility), approximately 30 yards
- Crochet hook, size H/8 (5.0mm) or your preferred size for this yarn
- Pen and Training Log (template below)
- Graph paper or blank paper for plotting your training curve
- Ruler
- Colored pencils or markers (2 colors)
- Timer or clock
- Scissors

Choosing Your New Stitch

Pick ONE stitch that you have never successfully executed before. The stitch should be genuinely new to you — something that will produce real errors for you to track. Some options, ranked roughly from less to more challenging:

- **Front Post Double Crochet (FPdc)** — if you have never worked post stitches
- **Back Post Double Crochet (BPdc)** — same family, different feel
- **Puff Stitch** — multiple yarn-overs pulled through together
- **Bullion Stitch** — many wraps around the hook, pulled through at once (quite challenging)
- **Linked Double Crochet** — inserting through the side of the previous stitch
- **Crocodile Stitch** — a multi-step stitch that creates scale-like texture

If you are an experienced crocheter who knows all common stitches, choose a technique instead: Tunisian crochet, broomstick lace, hairpin lace, or working with an unfamiliar hand position (e.g., knife grip if you normally use pencil grip).

Write down your chosen stitch: ____

Training Log Template

Row	Total Stitches Attempted	Errors (count)	Error Rate (errors/total)	Tension Rating (1-5)	Time to Complete Row	Notes on What Went Wrong
1						
2						
3						
4						
5						

Error types to track: missed insertion, wrong number of wraps, uneven tension, dropped loop, twisted stitch, stitch that had to be redone, stitch that looks visibly different from its neighbors.

Tension rating scale: 1 = very uneven / inconsistent, 2 = mostly uneven, 3 = mixed, 4 = mostly even, 5 = very consistent and even.

Steps

Part 1: Baseline and First Attempt (10 minutes)

1. Look up your chosen stitch in a crochet reference (book, video tutorial, or written instructions). Read or watch the instructions carefully, but do not practice yet. Just study.
2. Chain 16 (or an appropriate foundation for your chosen stitch — adjust if your stitch requires a specific chain multiple).
3. **Row 1:** Attempt your new stitch across the row. Work slowly and deliberately. Count every stitch attempted and every error. An "error" is any stitch where you had to redo it, where you did not follow the correct steps, or where the result is visibly wrong.
4. **Record** in your Training Log: total stitches, error count, error rate, tension rating (be honest), time to complete the row, and notes on what specifically went wrong.
5. **Do not frog this row.** Leave the errors in place. They are your training data.

Part 2: Training Epochs 2-5 (25 minutes)

1. **Row 2:** Turn your work (or continue, depending on the stitch). Attempt the same stitch across the row. Focus on correcting the specific errors you made in Row 1.
2. **Record** all metrics in your Training Log.
3. **Row 3:** Same stitch, same focus. Notice: are you thinking less about each sub-step? Are your hands starting to find the rhythm?
4. **Record.**
5. **Row 4:** By now, you may notice a shift — the stitch is starting to feel more natural. Some sub-actions (the insertion, the wrap) may be becoming automatic while others still require conscious thought.
6. **Record.**
7. **Row 5:** Your final training epoch. Give it your best effort. Try to execute each stitch smoothly and consistently.
8. **Record.**

Part 3: Plotting Your Training Curve (10 minutes)

1. On your graph paper, draw two axes:
 - Horizontal axis: Row number (1 through 5)
 - Vertical axis: Error rate (errors / total stitches)
2. Plot your error rate for each row. Connect the points with a line.
3. On the same graph (using a different color), plot your tension rating for each row.
4. **Label your graph:** "Training Curve: [Your Stitch Name]"
5. Look at the shape of your curve. Where was the steepest improvement? Where did it level off? Does it look like a typical neural network training curve (steep early drop, then gradual leveling)?

Part 4: Comparing to Neural Network Training (10 minutes)

1. On a separate section of your paper (or on the back), sketch a **typical neural network training curve**: a line that starts high (high loss), drops steeply in the early epochs, then gradually levels off toward a low value (but rarely reaches zero).
2. Compare your crochet training curve to this typical shape. Answer the following in writing:
 - Did your error rate follow a similar pattern — steep early improvement, then leveling off?
 - If your curve looks different (e.g., a sudden drop in Row 3 instead of a gradual decline), what might explain the difference? (A sudden "aha moment" when the stitch clicks? A particularly bad Row 2 when you tried something different?)
 - Did your tension rating improve along a similar curve, or did it follow a different pattern?
3. Write a brief paragraph (3-5 sentences) reflecting on which "weights" in your motor system adjusted the most during this exercise. Was it grip? Insertion depth? Wrap technique? Pull-through force? Tension control?

Part 5: Testing for Overfitting (10 minutes)

1. Now put down your practice swatch. Without looking at instructions, try to do 3 stitches of your new stitch into a fresh chain. Can you do it from memory?

2. Next, try mixing your new stitch with a stitch you already know well (e.g., alternate: new stitch, sc, new stitch, sc). How does switching between stitches feel? Does your rhythm break?
3. **Record** your observations:
 - Can you execute the new stitch without instructions? (If yes, some generalization has occurred.)
 - Is it harder when mixed with other stitches? (If yes, your motor system may have "overfit" to the repetitive practice context — it works when you do nothing but the new stitch, but falters when the context changes.)

Part 6: Reflection and Discussion (10 minutes)

1. Look at your finished swatch from Row 1 to Row 5. You can literally see your training curve in the fabric — the uneven early rows and the (hopefully) smoother later rows. This is a physical record of learning.
2. Discuss or journal:

Discussion Questions

- Where on your training curve did the biggest improvements happen — between which rows? What changed in your technique at that point?
- Did you experience a plateau (a row where your error rate did not improve much)? If so, what was happening during that plateau — were you consolidating earlier gains, or were you stuck?
- When did the stitch start to feel "automatic" — like you did not have to think through each step? In neural network terms, when did the weights start to converge?
- Compare your experience to the overfitting/generalization tradeoff. After 5 rows of the same stitch, how well could you use it in a new context (mixed with other stitches)? Were you "overfit" to the repetitive practice format?
- If you were to design a "training curriculum" for this stitch — a sequence of exercises to build skill most efficiently — what would you change about the order or structure of what you just did?

- How does the experience of learning this stitch (high error, gradual correction, eventual fluency) compare to the active inference framework: the crocheter predicts the stitch outcome, observes the error, and adjusts the action?

Wrap-Up

You have generated your own training data, plotted your own learning curve, and compared your biological learning process to neural network training. Keep your practice swatch — it is a physical artifact of your training epochs, with the early errors preserved right alongside the later improvements. In the next module, we explore how crocheters communicate and teach — how skill and knowledge flow through the network of a crochet circle.